Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A wave travelling on a string is described by $y(x,t) = 0.005\sin(80.0x - 3.0t)$
The period of the wave is

(A) 3.00 s

(B) 2.09 s

(C) 0.48 s

(D) 0.05 s

Q.2

A transverse wave is represented by $y = A$ sin $(\omega t - kx)$. For what value of the wavelength is the wave velocity equal to the maximum particle velocity?

(A) $\frac{\pi A}{2}$

(B) πA

(C) 2πA

(D) A

Q.3

The displacement y of a particle in a medium can be expressed as, $y = 10^{-6}\sin\left(100t + 20x + \frac{\pi}{4}\right) m$ where t is in
second and x in meter. The speed of the wave is

(A) 20 m/s

(B) 5 m/s

(C) 2000 m/s

(D) 5πm/s

Q.4

An earthquake generates both transverse (S) and longitudinal (P) sound waves in the earth. The speed of S waves in about 4.5 km/s and that of P waves is about 8.0 km/s. A seismograph records P and S waves from an earthquake. The first P wave arrives 4.0 min. before the first S wave. The epicenter of the earthquake is located at a distance about

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(A) $25\ \text{km}$

(B) $250\ \text{km}$

(C) $2500\ \text{km}$

(D) $5000\ \text{km}$

Q.5

A progressive wave is travelling in a medium such that frequency of oscillation and displacement amplitude of the particles of the medium are f and A respectively. The ratio of their acceleration amplitude and velocity amplitude is

(A) $2\pi f$

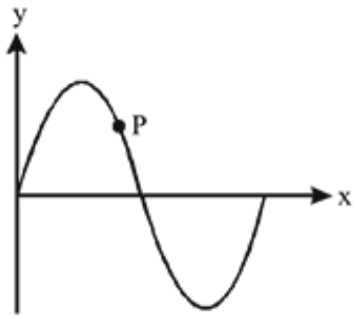
(B) πf

(C) $\frac{2\pi f}{A}$

(D) $\frac{\pi f}{A}$

Q.6

A transverse sinusoidal wave moves along a string in the positive x -direction at a speed of $10\ \text{cm/s}$. The wavelength of the wave is $0.5\ \text{m}$ and its amplitude is $10\ \text{cm}$. At a particular time t, the snap shot of thewave is shown in figure. The velocity ofpoint
P when its displacement is $5\ \text{cm}$ is



(A) $\frac{\sqrt{3}\pi}{50}\hat{j}\text{m/s}$

(B) $-\frac{\sqrt{3}\pi}{50}\hat{j}\text{m/s}$

(C) $\frac{\sqrt{3}\pi}{50}\hat{i}\text{m/s}$

(D) $-\frac{\sqrt{3}\pi}{50}\hat{i}\text{m/s}$

Q.7

Velocity of sound waves in air is $330\ \text{m/s}$. For a particular sound wave in air, a path difference of $40\ \text{cm}$ is equivalent to phase difference of 1.6π. The frequency of this wave is

(A) $165\ \text{Hz}$

(B) $150\ \text{Hz}$

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(C) $660\ \text{Hz}$

(D) $330\ \text{Hz}$

Q.8

A hospital uses an ultrasonic scanner to locate tumours in a tissue. The operating frequency of the scanner is 4.2MHz. The speed of sound in a tissue is $1.7\ \text{km/s}$. The wavelength of sound in tissue is close to

(A) $4 \times 10^{-4}\ \text{m}$

(B) $8 \times 10^{-4}\ \text{m}$

(C) $4 \times 10^{-3}\ \text{m}$

(D) $8 \times 10^{-3}\ \text{m}$

Q.9

A wave travelling along the x-axis is described by the equation $y(x,t) = 0.005\cos(\alpha x - \beta t)$
If the wavelength and the time period of the wave are $0.08\ \text{m}$ and $2.0\ \text{s}$, respectively, then
α and β in appropriate units are

(A) $\alpha = \frac{0.08}{\pi}, \beta = \frac{20}{\pi}$

(B) $\alpha = \frac{0.04}{\pi}, \beta = \frac{1.0}{\pi}$

(C) $\alpha = 12.50\pi, \beta = \frac{\pi}{20}$

(D) $\alpha = 25.00\pi, \beta = \pi$

Q.10

Consider the three waves z_1, z_2 and z_3 as $z_1 = A\sin(kx - \omega t); z_2 = A\sin(kx + \omega t)$
$z_3 = A\sin(ky - \omega t)$
Which of the following represents a standing wave?

(A) $\text{z}_1 + \text{z}_2$

(B) $\text{z}_2 + \text{z}_3$

(C) $\text{z}_3 + \text{z}_1$

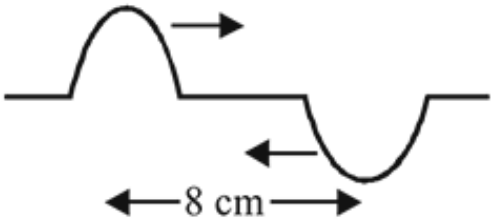
(D) $z_1 + z_2 + z_3$

Q.11

Two pulses in a stretched string whose centres are initially $8\ \text{cm}$ apart are moving towards each other as

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shown in the figure. The speed of each pulse is $2\ \mathrm{cm/s}$. After $2\ \mathrm{s}$, the total energy of the pulses will be



(A) Zero

(B) Purely kinetic

(C) Purely potential

(D) Partly kinetic and partly potential

Q.12

In a standing wave formed as a result of reflection from a surface, the ratio of the amplitude at an antinode to that at node is x. The fraction of energy that is reflected is

(A) $\left[\frac{x-1}{x}\right]^2$

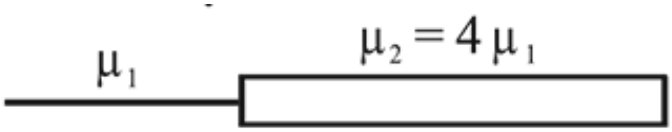
(B) $\left[\frac{x}{x+1}\right]^2$

(C) $\left[\frac{x-1}{x+1}\right]^2$

(D) $\left[\frac{1}{x}\right]^2$

Q.13

String 1 is connected with string 2 . The mass per unit length in string 1 is μ_1 and the mass per unit length in string 2 is $4\mu_1$. The tension in the strings is T. A travelling wave is coming from the left. What fraction of the energy in the incident wave goes into string 2 ?



(A) $1/8$

(B) $4/9$

(C) $2/3$

(D) $8/9$

Q.14

A composite string is made up by joining two strings of different masses per unit length μ and 4μ. The composite string is under the same tension. Atransverse wave pulse $Y = (6\ \mathrm{mm})\sin(5\,t + 40x)$, where $'t'$ is in seconds and $'x'$ is in metres, is sent along the lighter string towards the joint. The joint is at $x = 0$. The equation of the wave pulse reflected from the joint is

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(A) $Y = (2\ \text{mm})\sin(5t - 40x)$

(B) $Y = (4\ \text{mm})\sin(40x - 5t)$

(C) $Y = -(2\ \text{mm})\sin(5t - 40x)$

(D) $Y = (2\ \text{mm})\sin(5t - 10x)$

Q.15

Two travelling waves $y_1 = A\sin[k(x - ct)]$ and $y_2 = A\sin[k(x + ct)]$ are superimposed on string. The distance between adjacent nodes is

(A) ct/π

(B) $ct/2\pi$

(C) $\pi/2k$

(D) π/k

Answers & Solutions

Q.1 Answer:

$2.09\ \text{s}$

Solution:

Time period, $T = \frac{2\pi}{\omega}$**; from given eqn.** $\omega = 3.0\ \text{s}^{-1}$
or $\text{T} = \frac{2\pi}{3} = 2.09\ \text{s}$

Q.2 Answer:

$2\pi\text{A}$

Solution:

$y = A\sin(\omega t - kx)$
Particle velocity, $v_p = \frac{dy}{dt} = A\omega\cos(\omega t - kx)$
$\therefore \quad v_{p\,\text{max}} = A\omega$
wave velocity $= \frac{\omega}{k}$
$\therefore A\omega = \frac{\omega}{k}$
i.e., $A = \frac{1}{k}$ **But** $k = \frac{2\pi}{\lambda}$
- $\lambda = 2\pi A$

Q.3 Answer:

$5\ \text{m/s}$

Solution:

From equation, $\omega = 100 \therefore \frac{2\pi}{\text{T}} = 100 \Rightarrow \text{v} = \frac{100}{2\pi}$
$\frac{2\pi}{\lambda} = 20 \Rightarrow \lambda = \frac{2\pi}{20}$

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$\mathrm{v} = \lambda v = \frac{2\pi}{20} \times \frac{100}{2\pi} = 5\ \mathrm{m/s}$

Q.4 Answer:

$2500\ \mathrm{km}$

Solution:

Velocity of the S waves is $v_1 = 4.5\ \mathrm{km/s}$ the velocity of the P waves is $\mathrm{v}_2 = 8.0\ \mathrm{km/s}$.
Let the time taken by the S and P waves to reach the seismograph be t_1, and t_2 It is given that

$t_1 - t_2 = 4\ \mathrm{min} = 4 \times 60 = 240\ \mathrm{sec} \quad \ldots (i)$

Let the distance of the epicentre (km) be s.

Then

$s = v_1 t_1 = v_2 t_2$

$\Rightarrow 4.5 \times t_1 - 8 t_2 = 0 \Rightarrow t_2 = \frac{4.5}{8} t_1 \quad \ldots (ii)$

Using (i) and (ii)

$t_1 - t_2 = 240 \Rightarrow t_1 \left(1 - \frac{4.5}{8}\right) = 240$

$\Rightarrow t_1 = \frac{240 \times 8}{3.5} = 548.5\ \mathrm{s}.$

$\therefore \quad s = v_1 t_1 = 4.5 \times 548.5$

$= 2468.6 \approx 2500\ \mathrm{km}.$

Q.5 Answer:

$2\pi f$

Solution:

$v_{\max} = (2\pi f) A$

$a_{\max} = (2\pi f)^2 A \frac{a_{\max}}{v_{\max}} = (2\pi f)$

Q.6 Answer:

$\frac{\sqrt{3}\pi}{50} \hat{\mathrm{j}} \mathrm{m/s}$

Solution:

Since the wave is sinusoidal moving in positive x -axis the point will move parallel to y-axis therefore options (c) and (d) are ruled out. As the wave moves forward in positive X-direction, the point should move upwards i.e. in the positive Y-direction. Therefore correct option is a. Alternate solution-1:
Equation of a wave moving in positive x -axis is given as $y = A \sin(\omega t - \phi)$ or $v_P = A\omega \cos(\omega \mathrm{t} - \phi)$

Here $\mathrm{y} = 5\ \mathrm{cm},\ \mathrm{A} = 10\ \mathrm{cm}$

$\therefore 5 = 10 \sin(\omega \mathrm{t} - \phi) \Rightarrow \quad \omega \mathrm{t} - \phi = 30^\circ$

Substituting this value in the equation of velocity we get $\mathrm{v_P} = 0.10 \times \omega \cos 30^\circ$

Now $\mathrm{v} = v\lambda \therefore \mathrm{v} = \frac{\mathrm{v}}{\lambda} = \frac{0.10}{0.5} = 0.2$

$\therefore \omega = 2\pi v = 2\pi \times 0.2 = 0.4\pi$

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$\Rightarrow v_P = 0.1 \times 0.4\pi \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{50}\pi$

It has to be in positive y direction.

Q.7 Answer:

660 Hz

Solution:

Velocity of sound waves $= 330 \text{ m/s}$

$\Delta\phi = \text{K}\Delta\text{x}$

$\Delta\phi =$ phone difference

$\text{K} = \frac{2\pi}{\lambda}$

$\Delta\text{x} =$ **Path difference**

Putting the values given in question-

$1.6\pi = \frac{2\pi}{\lambda}\left(\frac{40}{100}\right)$

$\lambda = \frac{2\pi}{1.6\pi} \times \frac{40\times10}{100} = \frac{1}{2} \text{ m}$

Now,

$\text{V} = \text{f}\lambda$

$330 = \text{f}\left(\frac{1}{2}\right)$

$\text{f} = 660 \text{ Hz}$

Q.8 Answer:

$4 \times 10^{-4} \text{ m}$

Solution:

Speed of sound in the tissue, $v = 1.7 \text{ km/s} = 1.7 \times 10^3 \text{ m/s}$

Operating frequency of the scanner, $v = 4.2 \text{ MHz}$,

$= 4.2 \times 10^6 \text{ Hz}$

The wavelength of sound in the tissue is given as,

$\lambda = v/\text{v}$

$= 1.7 \times 10^3/4.2 \times 10^6$

$= 4.1 \times 10^{-4} \text{ m}$

Q.9 Answer:

$\alpha = 25.00\pi, \beta = \pi$

Solution:

$\alpha = \frac{2\pi}{0.08} = 25\pi$

$\beta = \frac{2\pi}{2} = \pi$

Q.10 Answer:

$\text{z}_1 + \text{z}_2$

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Solution:

Standing waves are produced when two waves propagate in opposite direction. As $z_1 \& z_2$ are propagating in $+$vex -axis $\&$
-ve x -axis
so, $z_1 + z_2$ will represent a standing wave.

Q.11 Answer:

Purely kinetic

Solution:

After 2 seconds, the pulses overlap each other completely and as a result, the string is straight and therefore does not have any potential energy and its entire energy must be kinetic energy.

Q.12 Answer:

$\left[\frac{x-1}{x+1}\right]^2$

Solution:

Subscript o, r stands for original and reflected wave respectively.

Given

$\frac{A_0+Ar}{A_0+A_r} = x$

Applying componendo and dividindo both sides

$\frac{A_0+Ar+A_0-A_r}{A_0+A_r-A_0+A_r} = \frac{x+1}{x-1}$

$A_r = \frac{x-1}{x+1} A_0$

so $A_r^2 = \left(\frac{x-1}{x+1}\right)^2 A_0^2$ **and** $E \propto A^2$

$E_r = \left(\frac{x-1}{x+1}\right)^2 E_0$

$\frac{E_r}{E_0} = \left(\frac{x-1}{x+1}\right)^2$

Q.13 Answer:

$8/9$

Solution:

Let v_1 and v_2 are the velocities of the waves in string 1 and string 2 respectively

$v_1 = \sqrt{\frac{T}{\mu_1}} \quad v_2 = \sqrt{\frac{T}{4\mu_1}} \quad v_1 = 2v_2$

Then the energy reflected in string 1 will be

$A_{\text{transmitted}} = \left(\frac{2v_2}{v_1+v_2}\right) A_{\text{incident}}$

$E_{\text{reflected}} = \frac{1}{3} E_{\text{incident}}$

$E_{\text{transfered}} = \frac{8}{9} E_{\text{incident}}$

Q.14 Answer:

$Y = -(2\text{ mm})\sin(5t - 40x)$

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Solution:

Since the wave is travelling from low to more dense medium, thus there will be inversion of the reflected wave . Since it travels back from origin, thus $40\mathrm{x}$ will become $-40\mathrm{x}$. Also, amplitude of reflected wave is given by:

$\mathrm{A_r} = \frac{\mathrm{Z_1}-\mathrm{Z_2}}{\mathrm{Z_1}+\mathrm{Z_2}}\ \mathrm{A}. \quad \mathrm{Z} = \mu \mathrm{c} = \mu\sqrt{\frac{\mathrm{T}}{\mu}} = \sqrt{\mu \mathrm{T}}$

Thus $\mathrm{A}_r = \frac{\sqrt{\mu \mathrm{T}}-\sqrt{4\mu \mathrm{T}}}{\sqrt{\mu \mathrm{T}}+\sqrt{4\mu \mathrm{T}}}\mathrm{A} = \frac{-1}{3}\ \mathrm{A}$

Thus it become $-6/3 = -2.$

Q.15 Answer:

π/k

Solution:

$y = A\sin(kx - kct) + A\sin(kx + kct)$

$= 2A\sin\left(\frac{kx-kct+kx+kct}{2}\right) \cdot \cos\left(\frac{kx-kct-kx-kct}{2}\right)$

$= 2A\sin(kct) \cdot \cos kx$

Thus $\frac{2\pi}{\lambda} = k, \therefore \lambda = \frac{2\pi}{k}$

The distance between adjacent nodes $= \lambda/2 = \pi/\mathrm{k}$

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